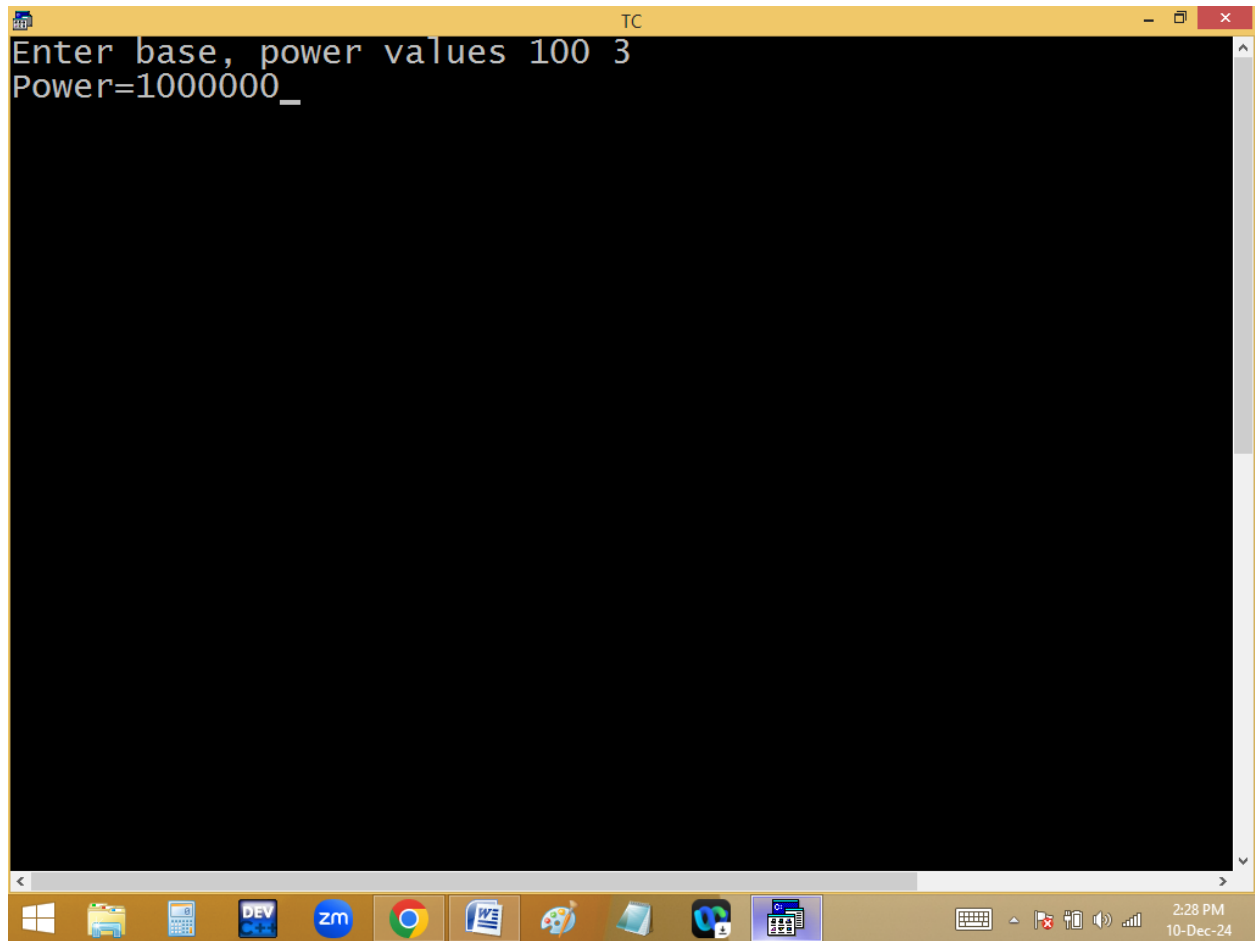


Finding power using a user defined function with arguments and return value:

```
TC
File Edit Run Compile Project Options Debug
Line 1 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
long power(int b, int p); /* fun declaration */
void main()
{
int b,p;;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
printf("Power=%ld",power(b,p)); /* fun calling */
getch();
}
long power(int b, int p) /* fun definition */
{
long r=1;
while(p>=1)
{
r=r*b;
p--;
}
return r;
}
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



```
TC
Enter base, power values 100 3
Power=1000000_
```

Finding power sum using a user defined function:

```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 23 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
long power(int b, int p); /* fun declaration */
void main()
{
int b,p;;
clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
printf("Power sum=%ld",power(b,p)); /* fun calling */
getch();
}
long power(int b, int p) /* fun definition */
{
long r=1,s=0;
while(p>=1)
{
r=r*b;s+=r;
p--;
}
return s;
}
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

```

Enter base, power values 2 3
Power sum=14_

```

```

long power(int b, int p)
{
    while( p >= 1 )
    {
        r = b * r; s += r;
        p--; ✓
    }
    return s;
}

```

$\frac{p}{3}$	$\frac{b}{2}$	$\frac{r}{1}$	$\frac{s}{0}$
2	2 ×	2 = 2 +	0 = 2
1	2 ×	4 = 2 +	2 = 6
Ø	2 ×	8 = 4 +	6 = 14

$$2^1 + 2^2 + 2^3 = 14$$

Using pow():

```
TC
File Edit Run Compile Project Options Debug
Line 7 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
#include<math.h>
long power(int b, int p); /* fun declaration */
void main()
{
int b,p; clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
printf("Power sum=%ld",power(b,p)); /* fun calling */
getch();
}
long power(int b, int p) /* fun definition */
{
long s=0;
while(p>=1)
{
s+=pow(b,p);
p--;
}
return s;
}
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

```

Enter base, power values 2 3
Power sum=14

```

```

long power(int b, int p)
{
    while( p >= 1 )
    {
        s += pow(b, p);
        p--; ✓
    }
    return s;
}

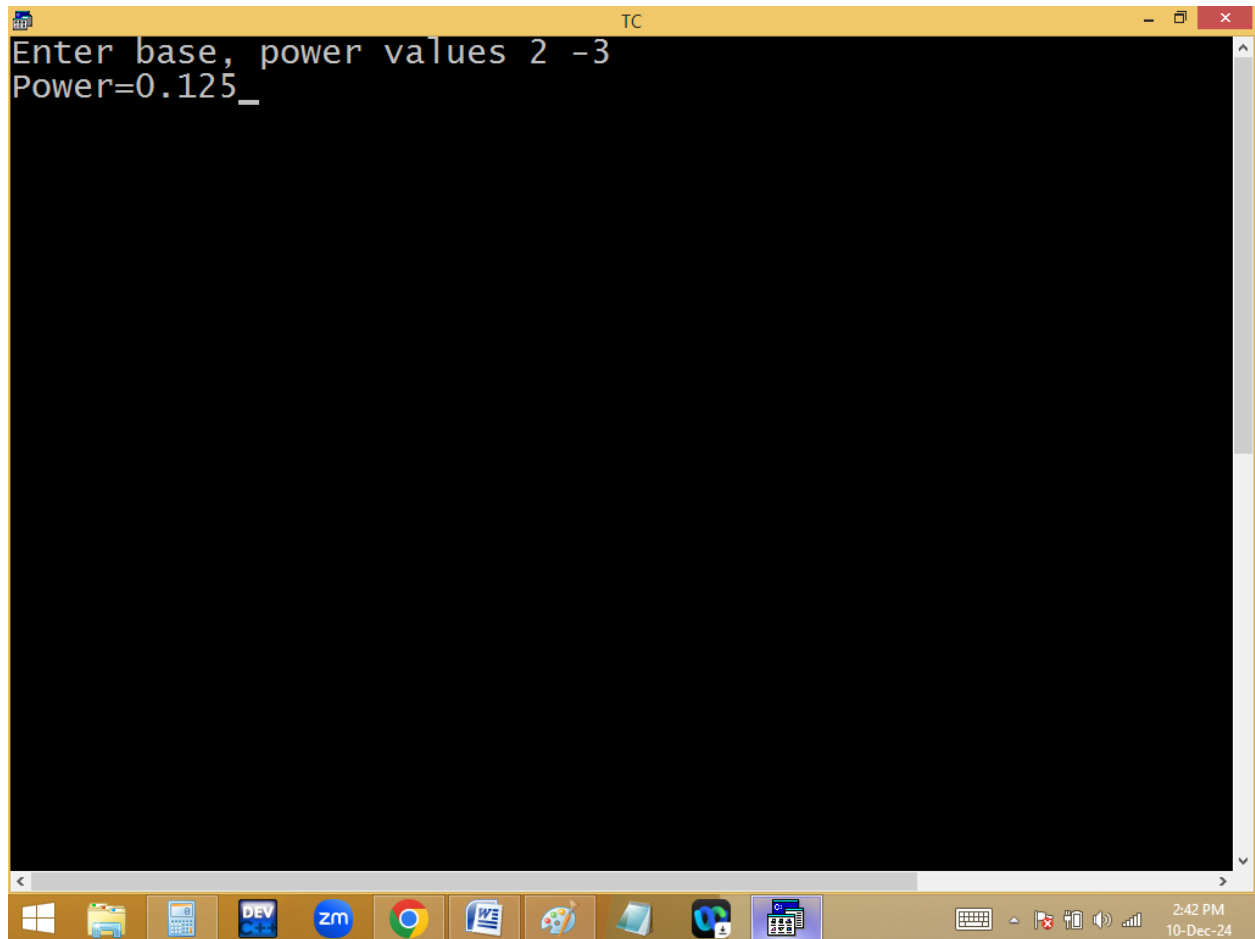
```

$\frac{p}{3}$	$\frac{b}{2}$
2	2
1	2
0	

$\frac{s}{0+8}$
4
2
14

$2^1 + 2^2 + 2^3 = 14$


```
TC
File Edit Run Compile Project Options Debug
Line 3 Col 6 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
float power(int b, int p); /* fun declaration */
void main()
{
int b,p; clrscr();
printf("Enter base, power values "); scanf("%d%d",&b,&p);
printf("Power=%.3f",power(b,p)); /* fun calling */
getch();
}
float power(int b, int p) /* fun definition */
{
float r=1; int t=p; if(p<0)p=-p;
while(p>=1)
{
r=r*b;;
p--;
}
if(t>=0) return r; else return 1/r;
}
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```

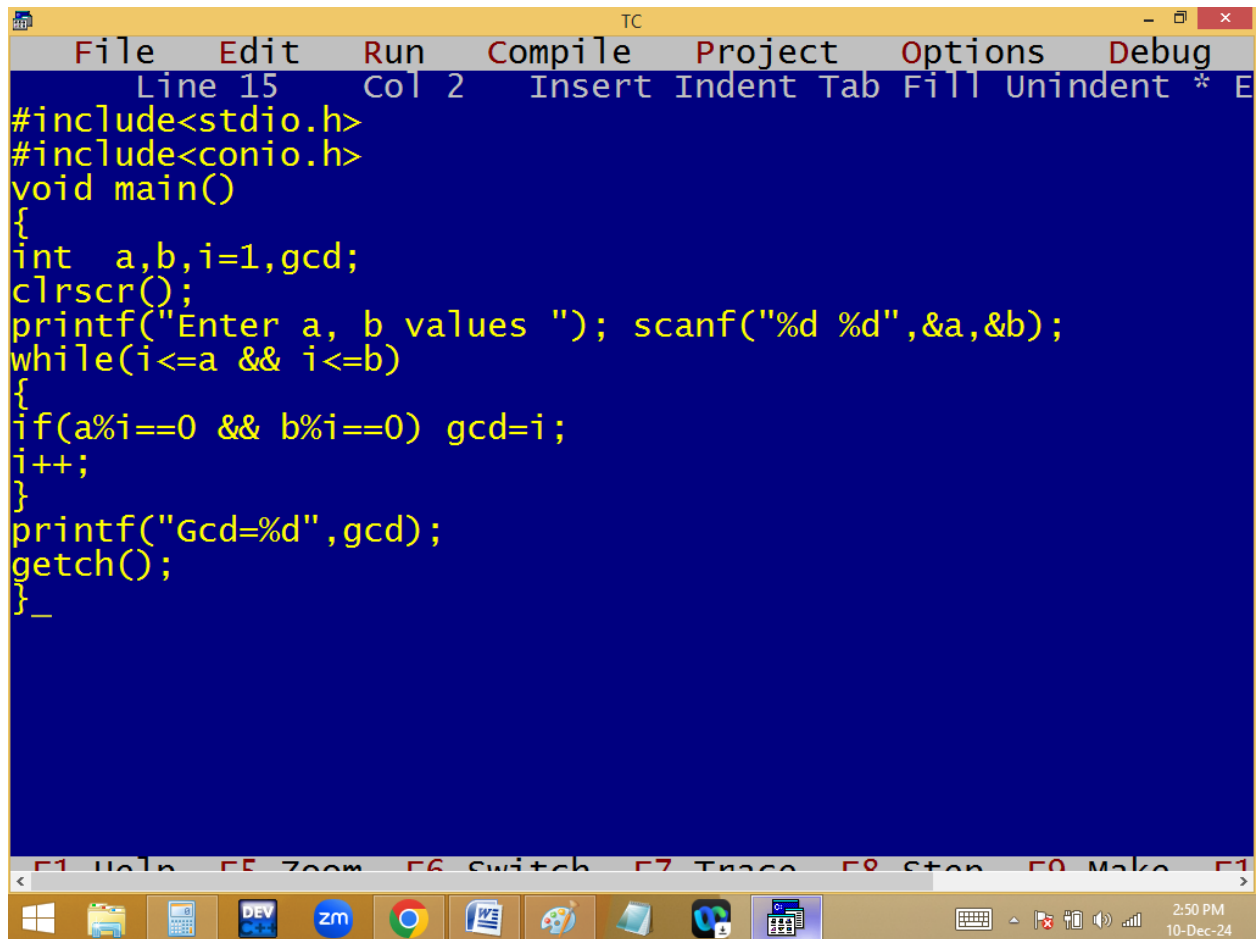


```
TC
Enter base, power values 2 -3
Power=0.125_
```

Finding gcd or hcf of given two numbers:

4 factors are 1, 2, 4

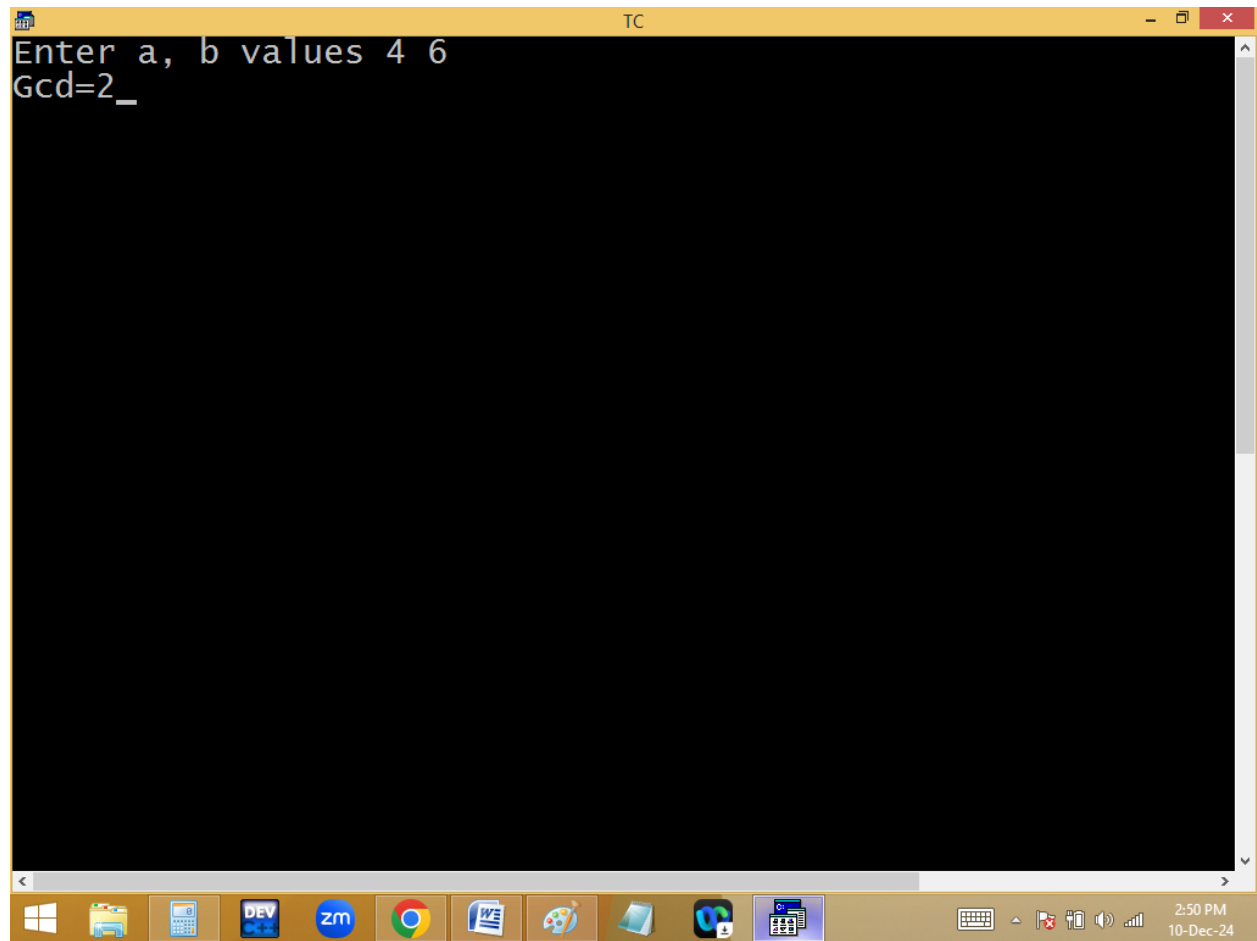
6 factors are 1, 2, 3, 6



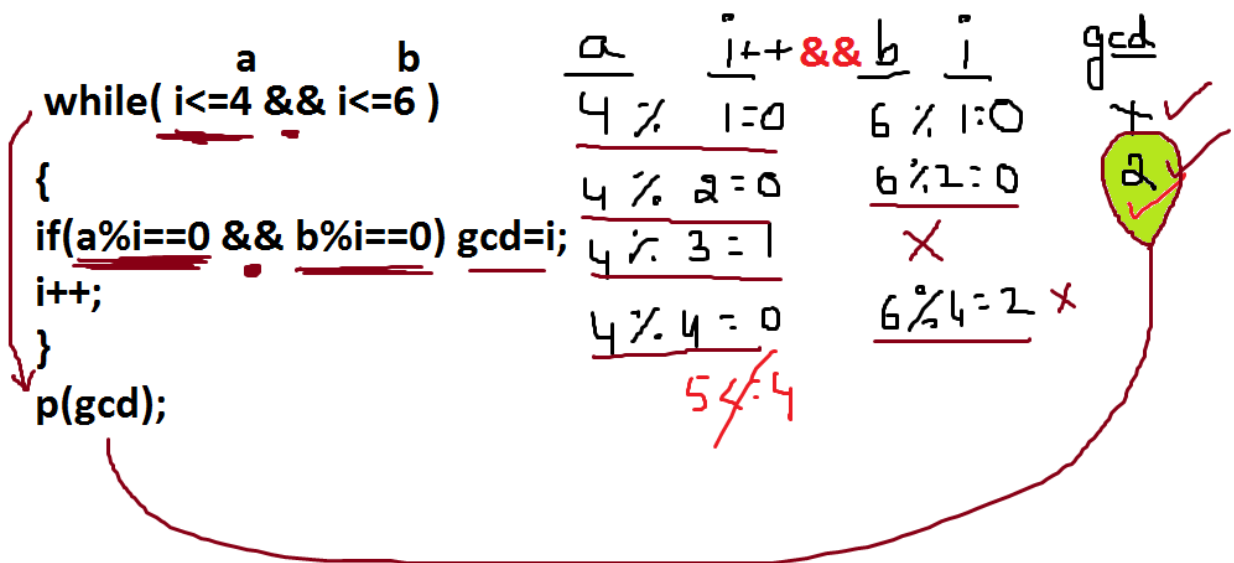
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 15", "Col 2", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains a C program for finding the GCD of two numbers. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,i=1,gcd;
    clrscr();
    printf("Enter a, b values "); scanf("%d %d",&a,&b);
    while(i<=a && i<=b)
    {
        if(a%i==0 && b%i==0) gcd=i;
        i++;
    }
    printf("Gcd=%d",gcd);
    getch();
}
```

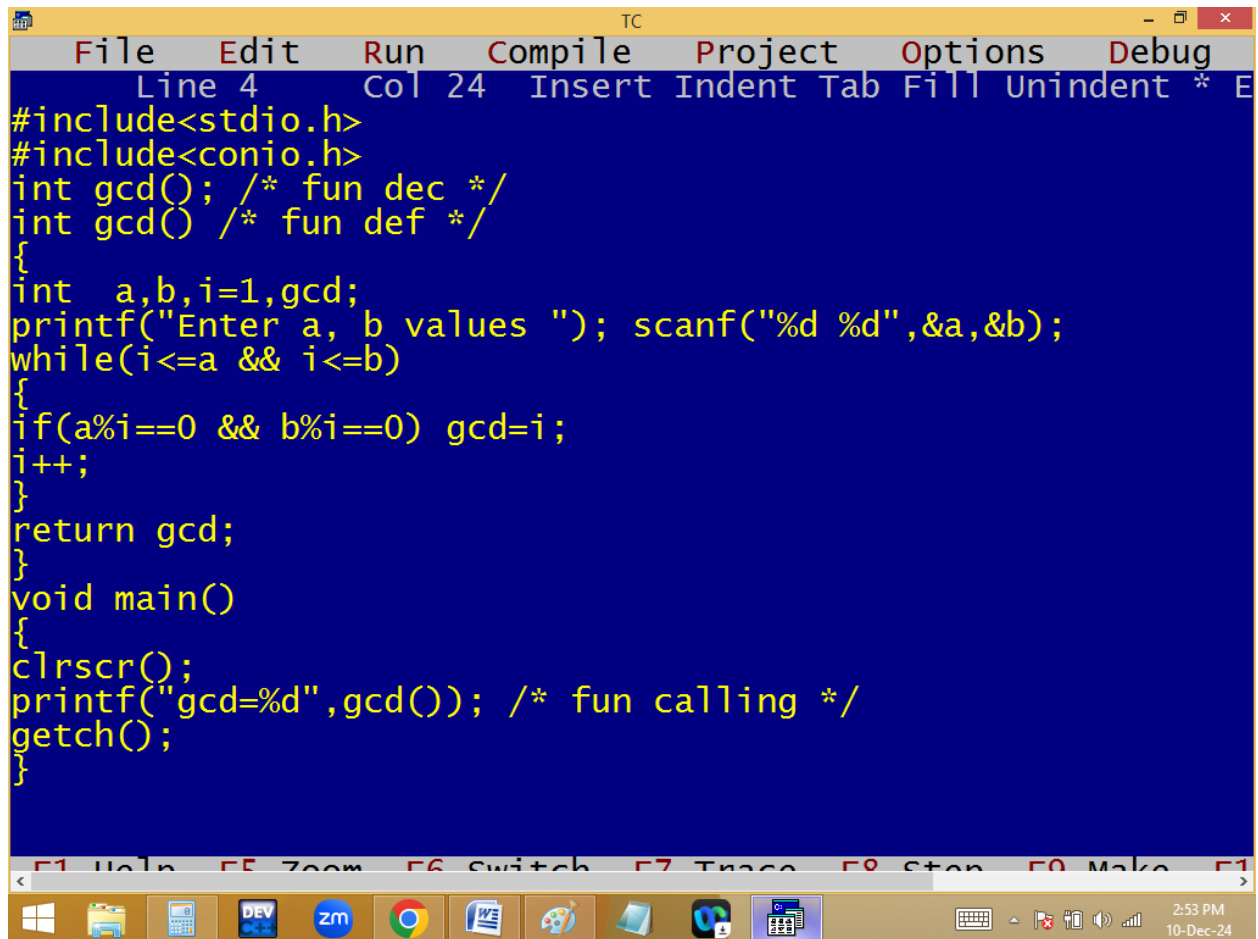
At the bottom of the window, there is a toolbar with icons for various functions, and a status bar on the far right showing the time "2:50 PM" and the date "10-Dec-24".



```
TC
Enter a, b values 10 15
Gcd=5_
```



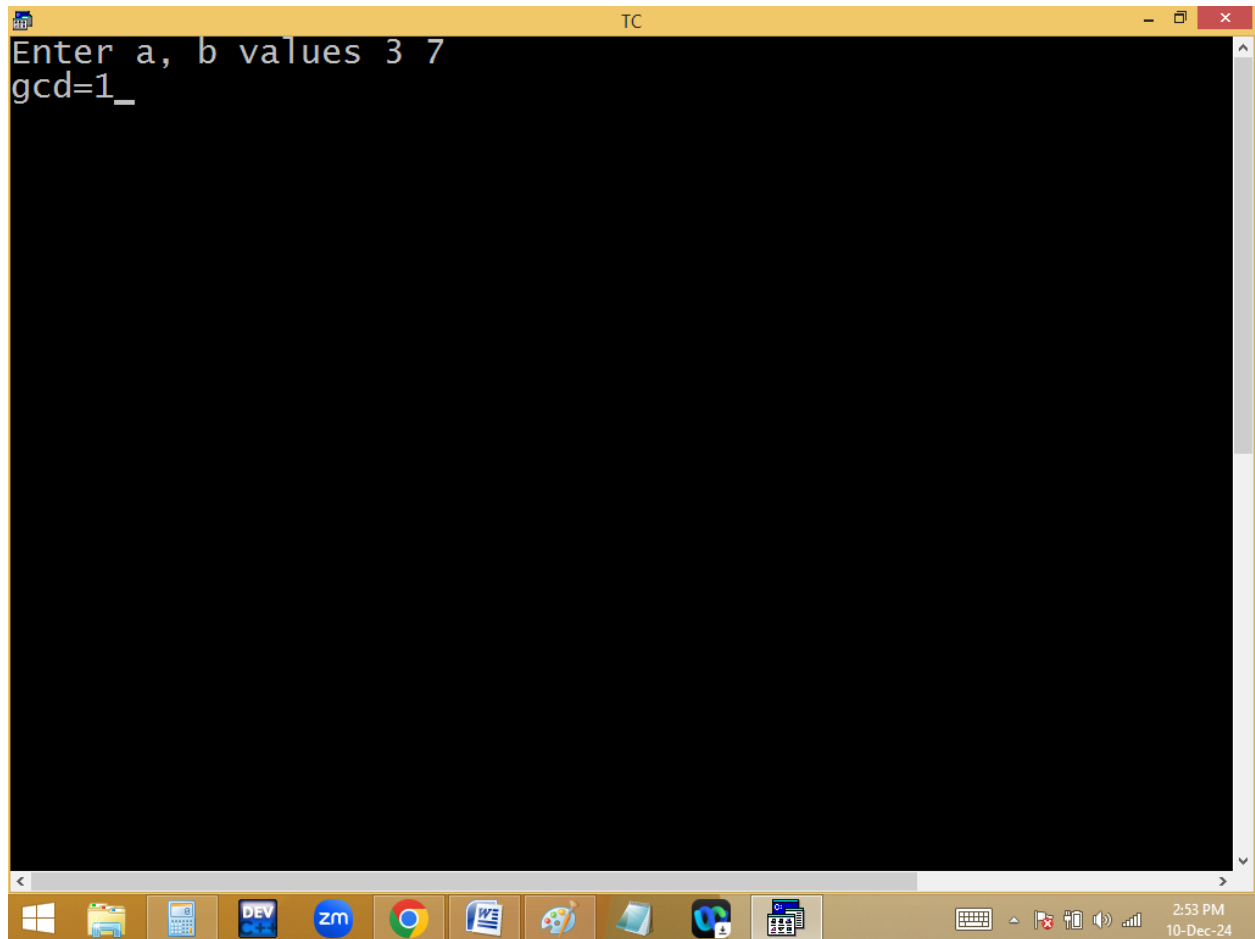
With udf:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 4", "Col 24", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
int gcd(); /* fun dec */
int gcd() /* fun def */
{
    int a,b,i=1,gcd;
    printf("Enter a, b values "); scanf("%d %d",&a,&b);
    while(i<=a && i<=b)
    {
        if(a%i==0 && b%i==0) gcd=i;
        i++;
    }
    return gcd;
}
void main()
{
    clrscr();
    printf("gcd=%d",gcd()); /* fun calling */
    getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar on the right shows the time "2:53 PM" and the date "10-Dec-24".



The screenshot shows a Turbo C++ (TC) window with a yellow title bar labeled 'TC'. The main window area is black, and the text 'Enter a, b values 3 7' and 'gcd=1_' is displayed in white. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:53 PM on 10-Dec-24.

```
Enter a, b values 3 7
gcd=1_
```

Finding lcm of given two numbers:

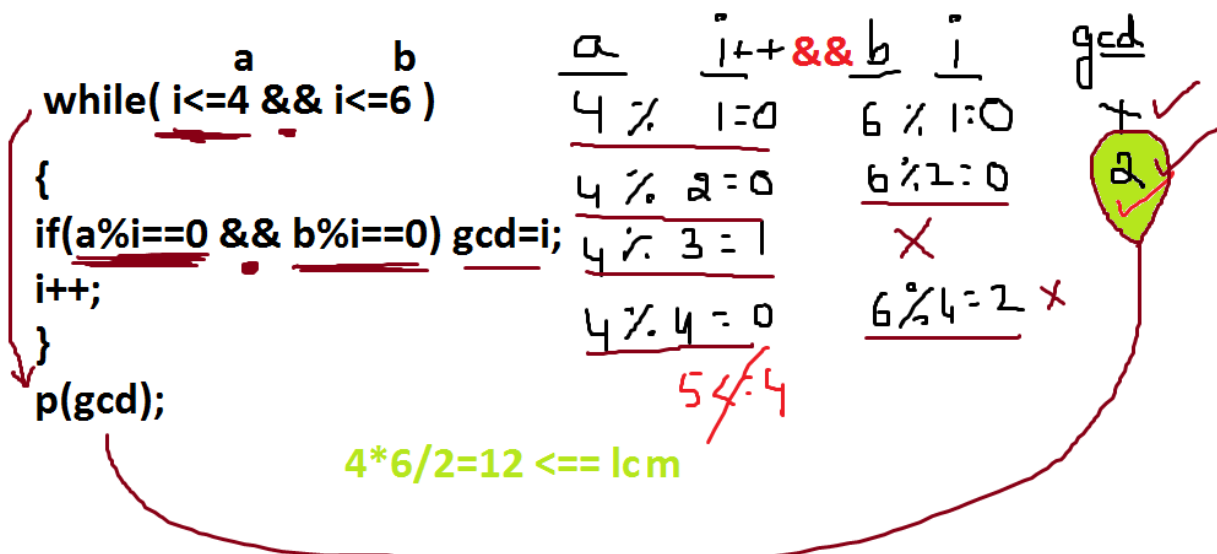
With gcd: $a * b / \text{gcd}$

```
TC
File Edit Run Compile Project Options Debug
Line 16 Col 18 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
int gcd(); /* fun dec */
int gcd() /* fun def */
{
int a,b,i=1,gcd;
printf("Enter a, b values "); scanf("%d %d",&a,&b);
while(i<=a && i<=b)
{
if(a%i==0 && b%i==0) gcd=i;
i++;
}
return a*b/gcd;
}
void main()
{
clrscr();
printf("lcm=%d",gcd()); /* fun calling */
getch();
}
```

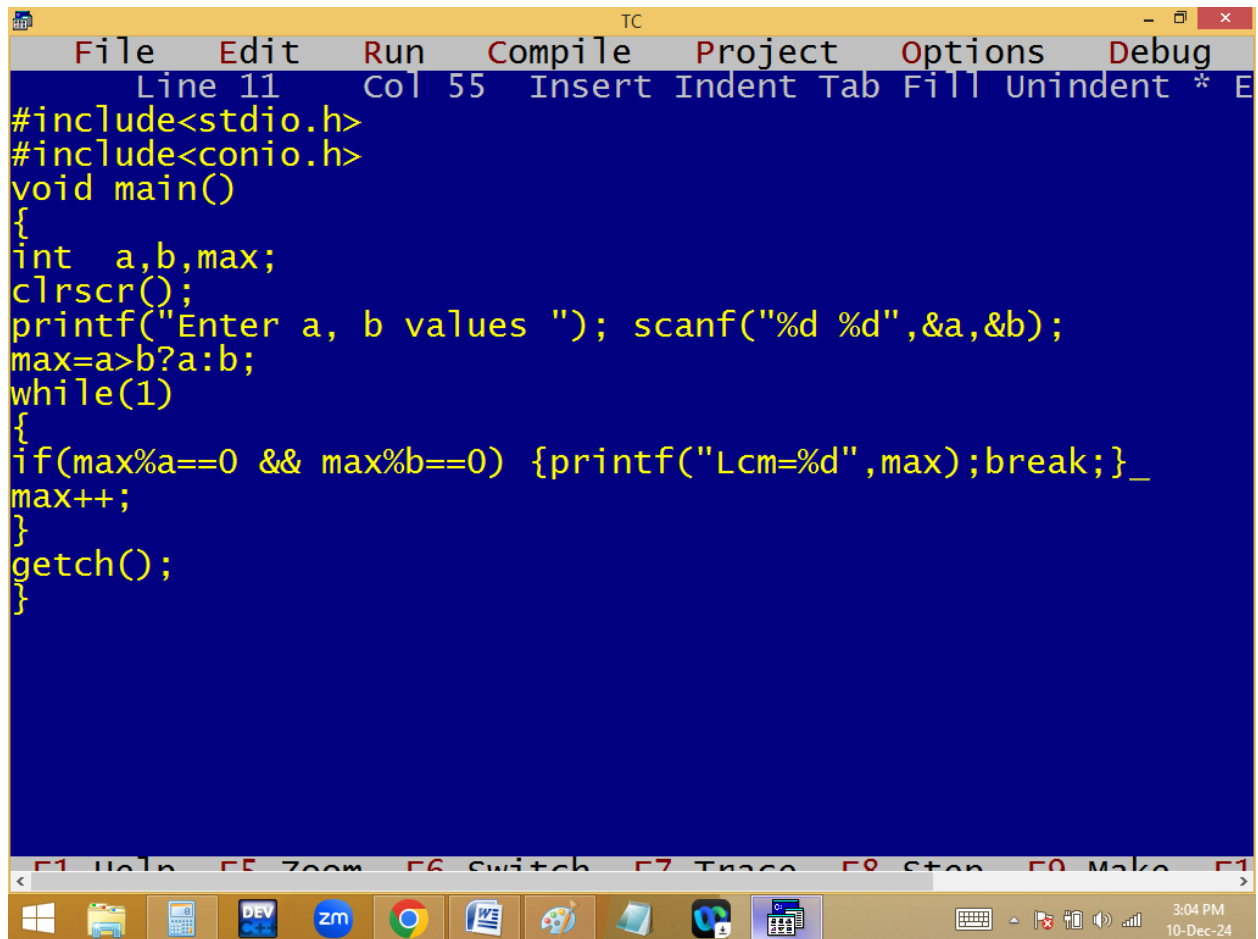
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

2:55 PM
10-Dec-24


```
TC
Enter a, b values 4 6
lcm=12_
```



Without gcd:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 11", "Col 55", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a,b,max;
    clrscr();
    printf("Enter a, b values "); scanf("%d %d",&a,&b);
    max=a>b?a:b;
    while(1)
    {
        if(max%a==0 && max%b==0) {printf("Lcm=%d",max);break;}_
        max++;
    }
    getch();
}
```

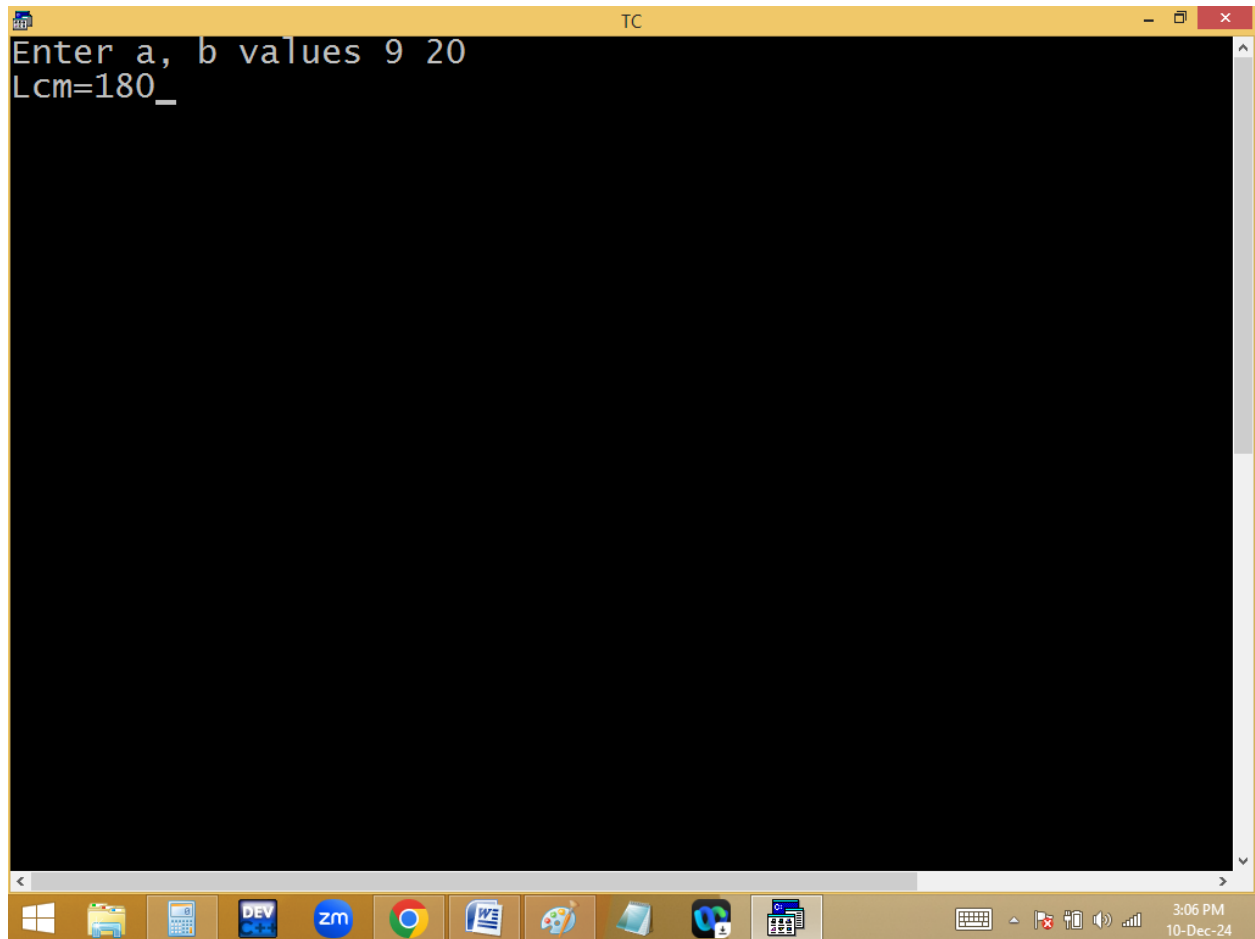
At the bottom of the window, there is a toolbar with icons for various functions and a status bar. The status bar on the right shows the time "3:04 PM" and the date "10-Dec-24".

```
TC
Enter a, b values 4 6
Lcm=12
```

`max=a>b?a:b;`

```
while(1)
{
if(max%a==0&&max%b==0)
{
p("Lcm=%d",max); break;
}
max++;
}
```

<u>max</u>	<u>a</u>	<u>max</u>	<u>b</u>
6 %	4 = 2	6	6
7 %	4 = 3		
8 %	4 = 2	8 %	6 = 2
9 %	4 = 1		
10 %	4 = 2		
11 %	4 = 3		
12 %	4 = 0	12 %	6 = 0



```
Enter a, b values 9 20
Lcm=180_
```

Finding digits sum:

123 → 1+2+3=6

$$\begin{array}{r} 11 \\ \hline 123 \end{array} \% 10 = 3 \quad + \quad \begin{array}{r} 5 \\ \hline 0 \end{array} = 3$$

$$123 / 10 = 12$$

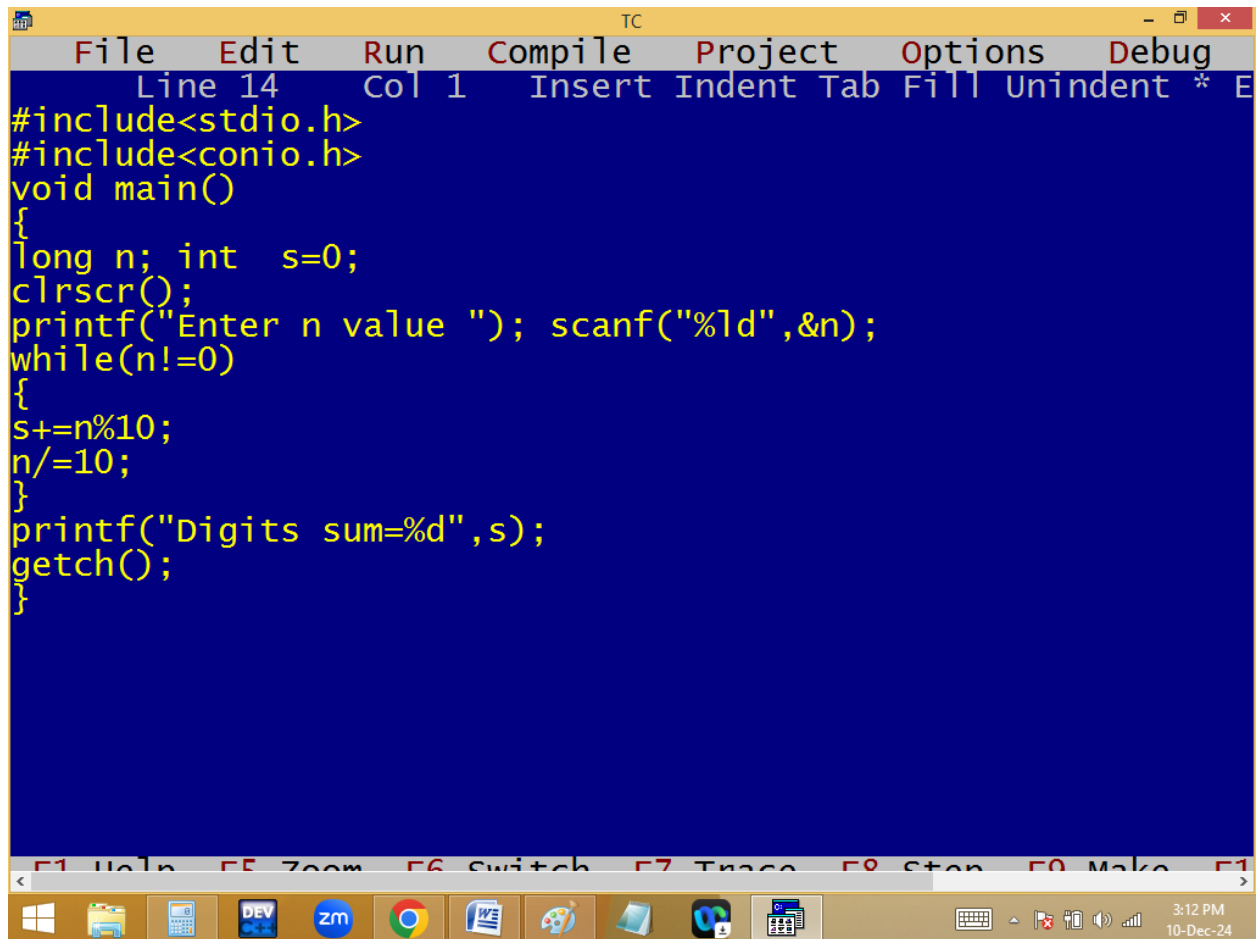
$$12 \% 10 = 2 \quad + \quad 3 = 5$$

$$12 / 10 = 1$$

$$1 \% 10 = 1 \quad + \quad 5 = 6$$

Any no%10 gives last digit.

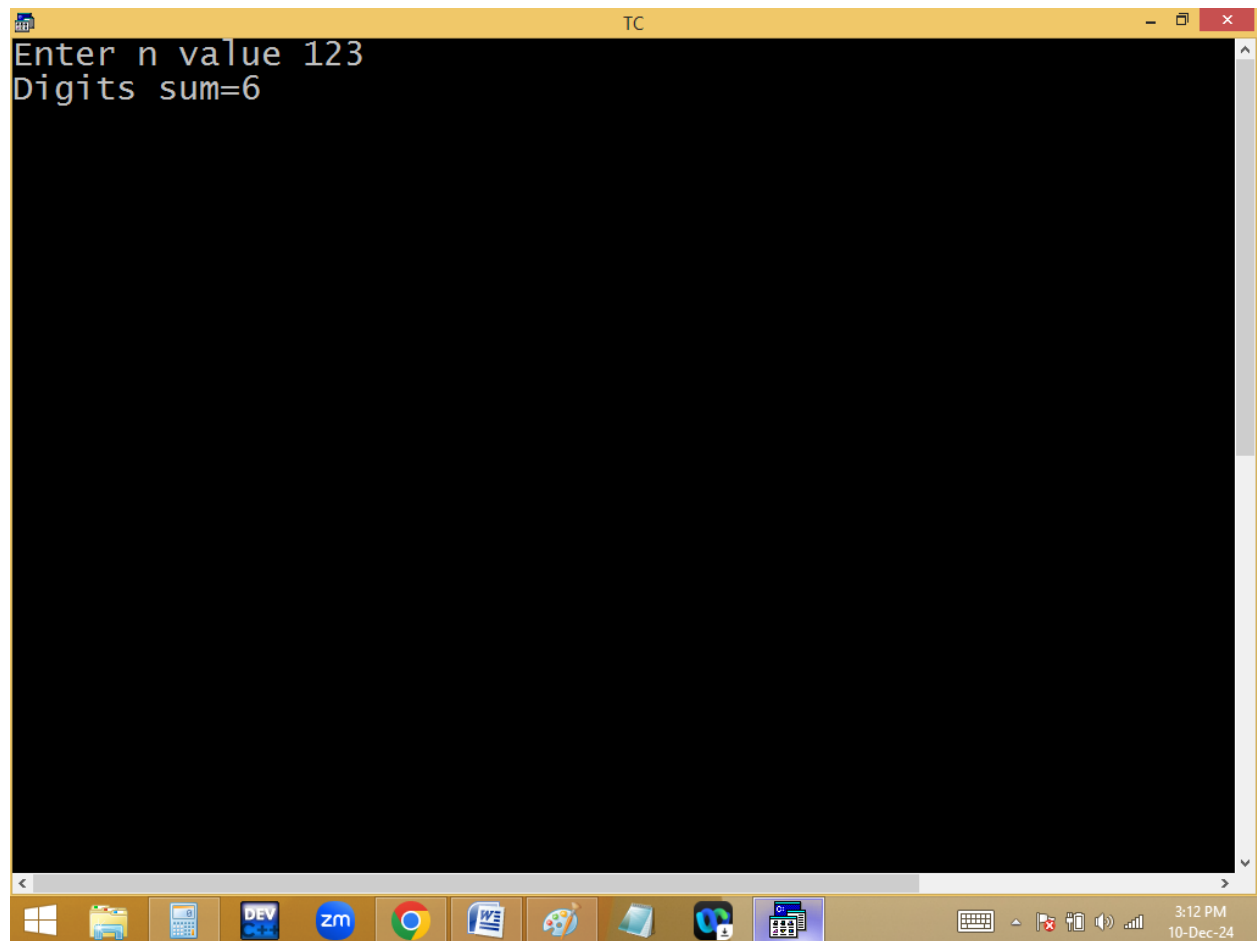
Any no/10 removes last digit.

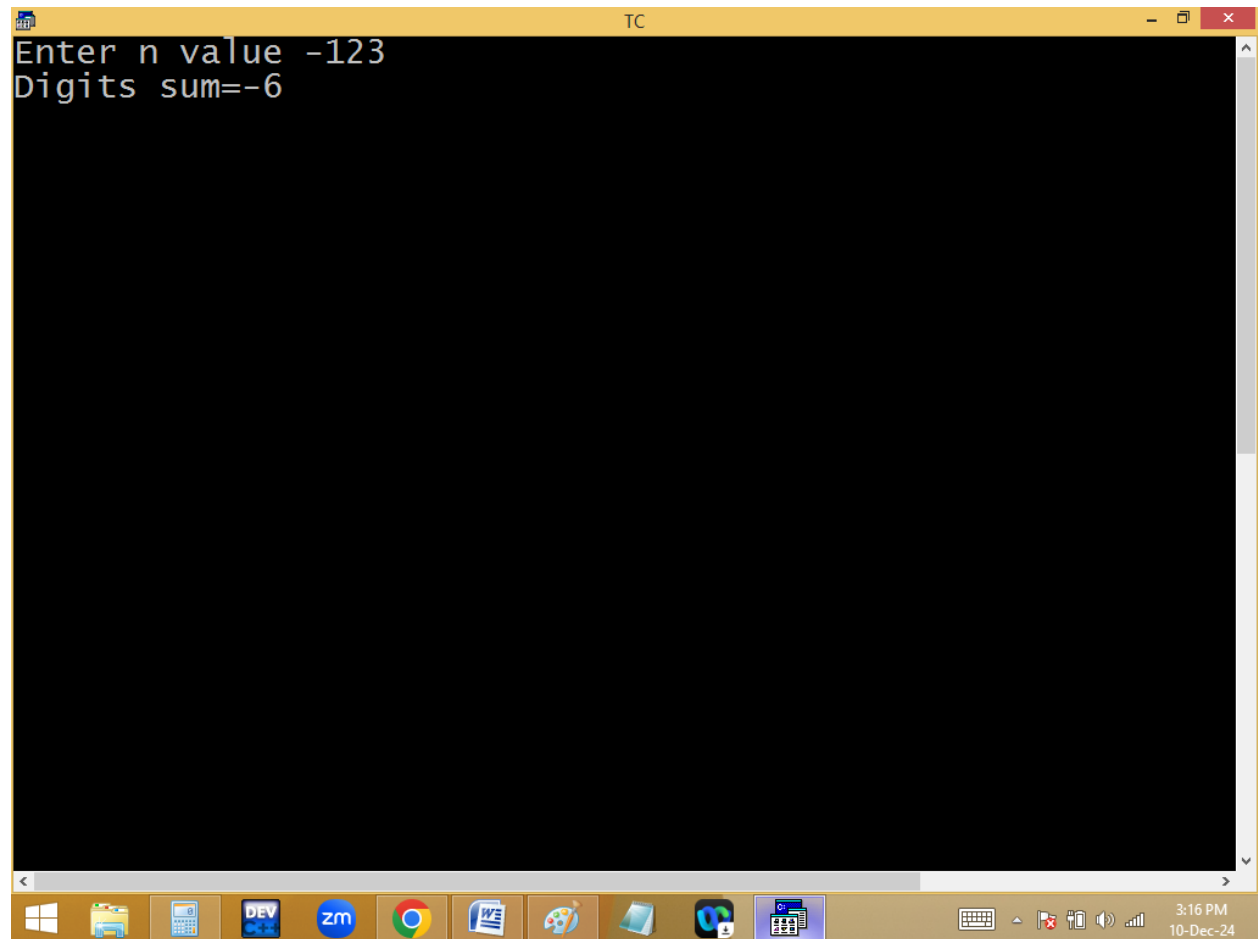


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 14", "Col 1", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

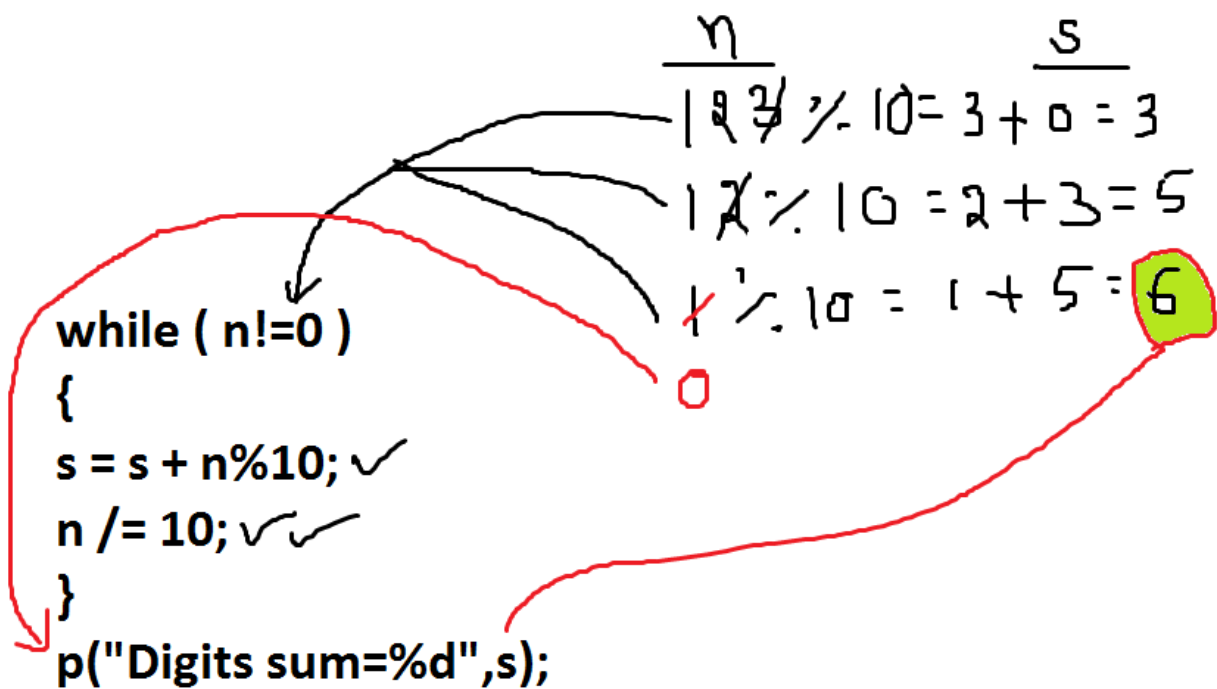
```
#include<stdio.h>
#include<conio.h>
void main()
{
    long n; int  s=0;
    clrscr();
    printf("Enter n value "); scanf("%ld",&n);
    while(n!=0)
    {
        s+=n%10;
        n/=10;
    }
    printf("Digits sum=%d",s);
    getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions, and a status bar showing the time "3:12 PM" and the date "10-Dec-24".



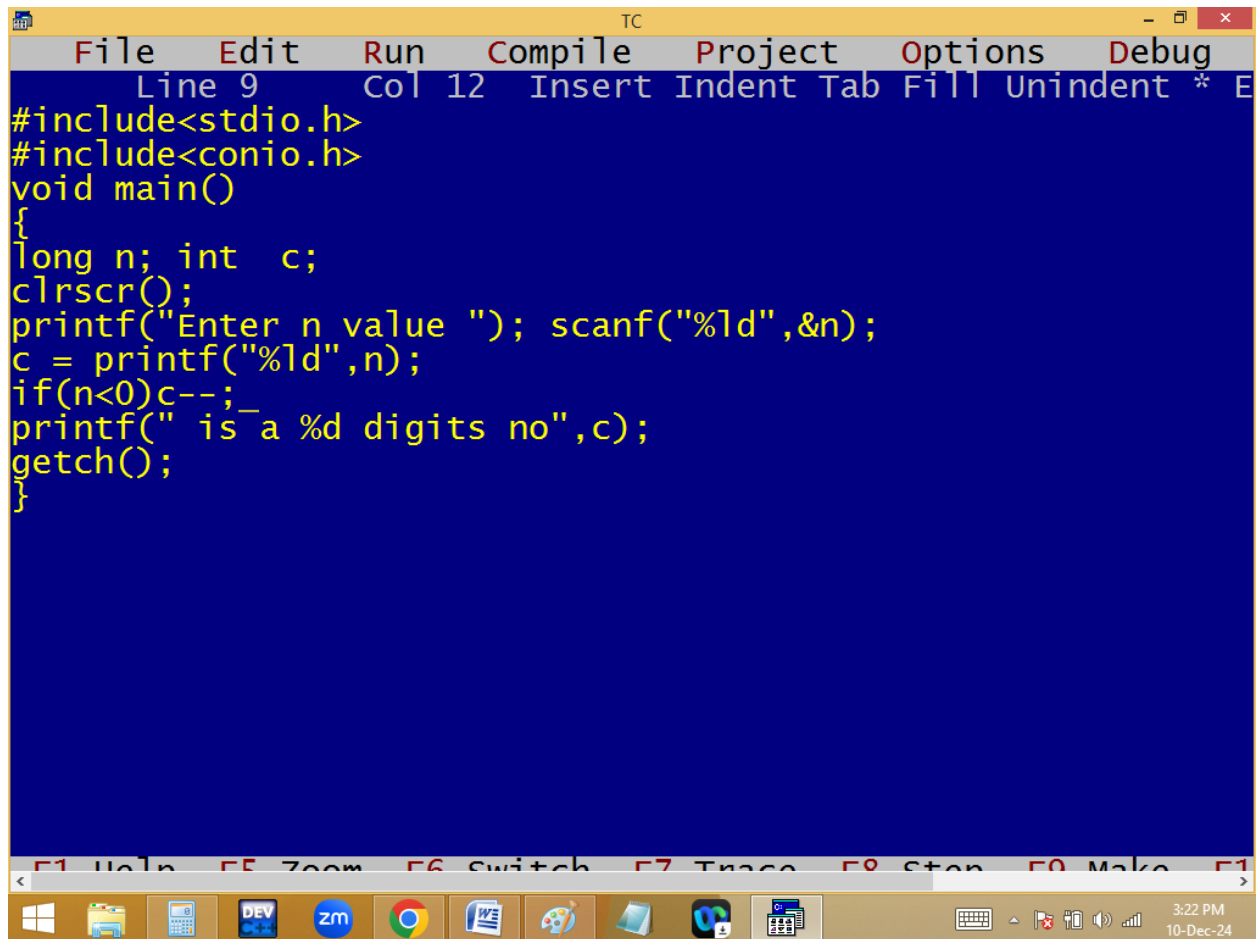



```
Enter n value 0
Digits sum=0_
```



**Finding no of digits in given no without using loop
/ goto label.**

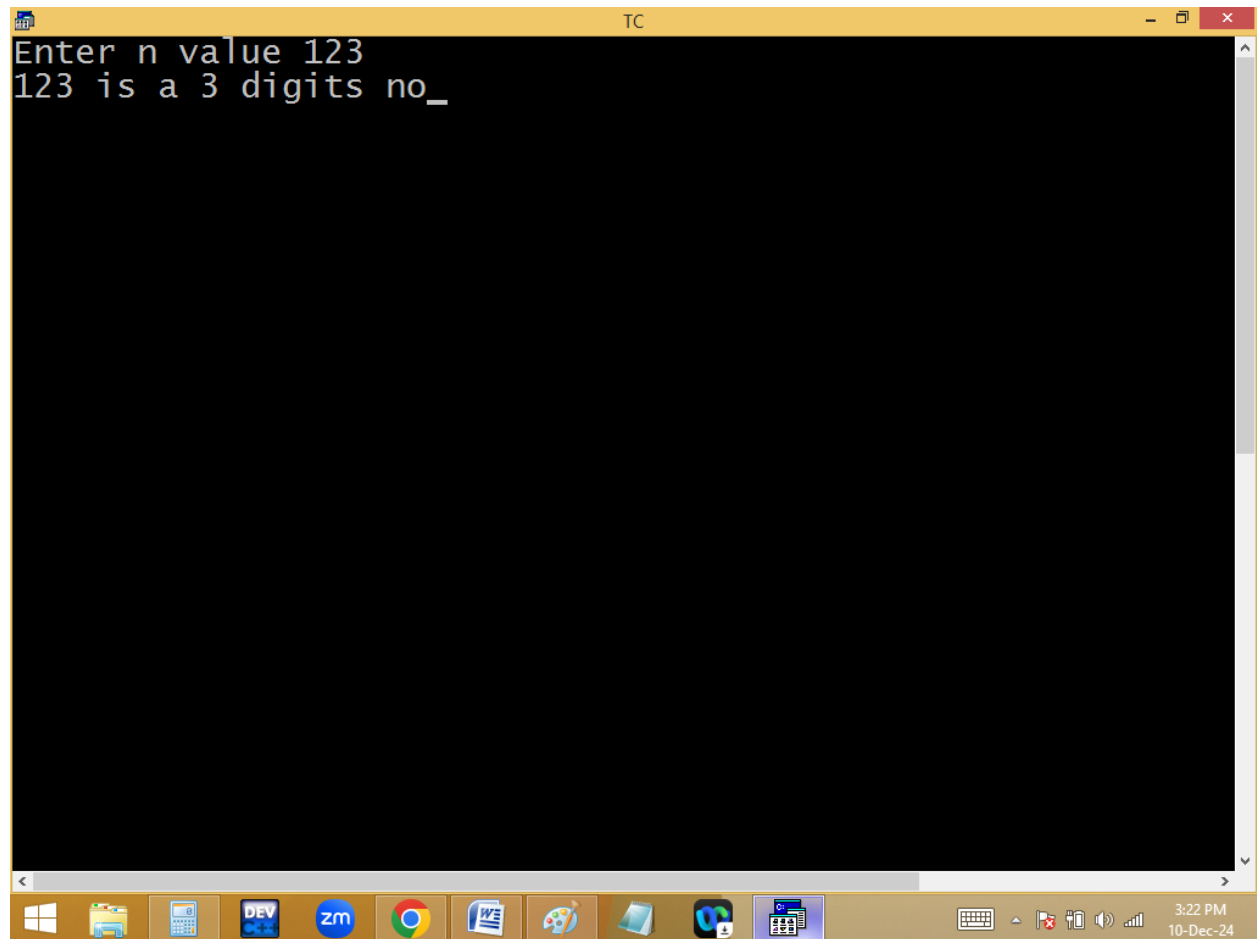
Eg: 143 → 3 digit no

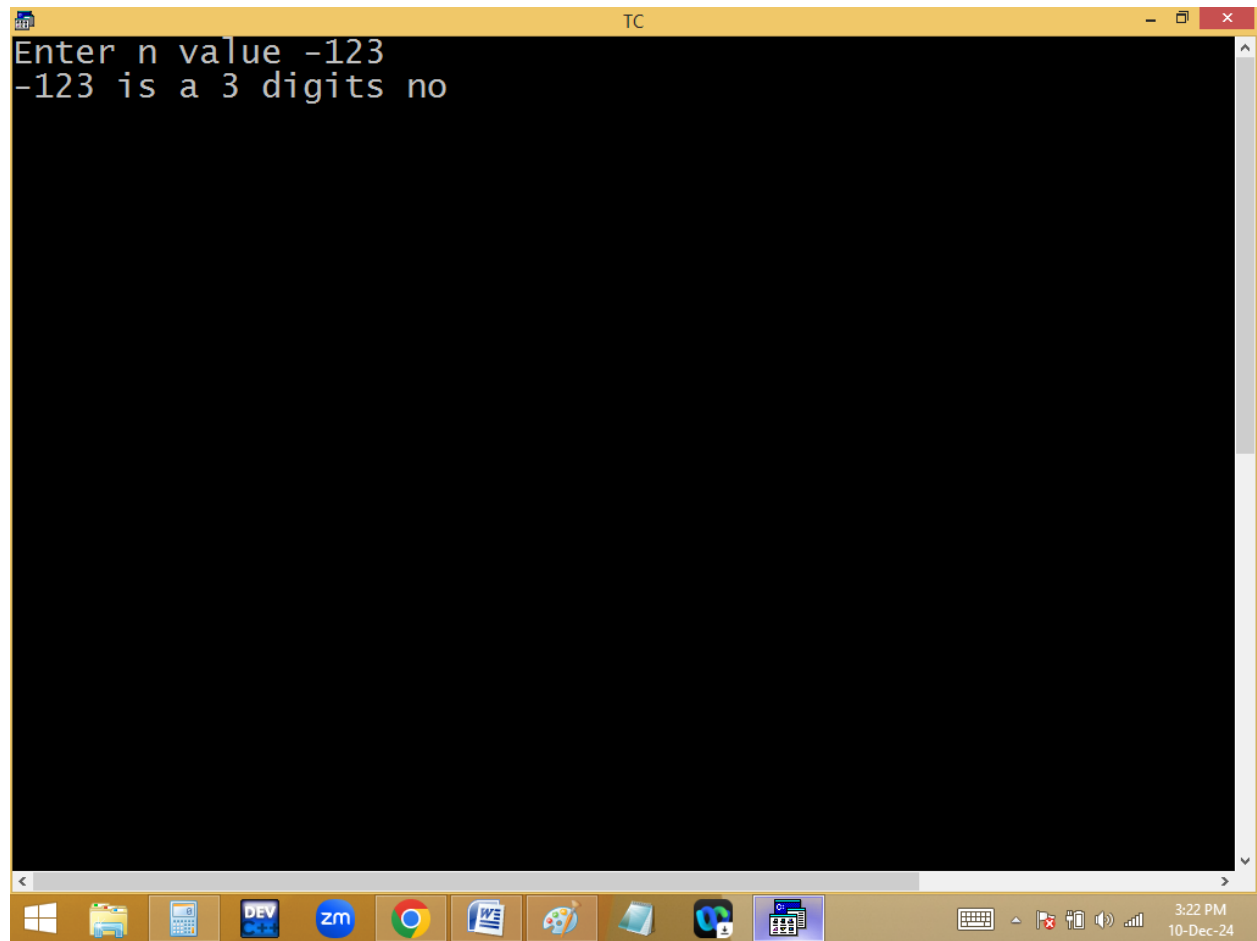


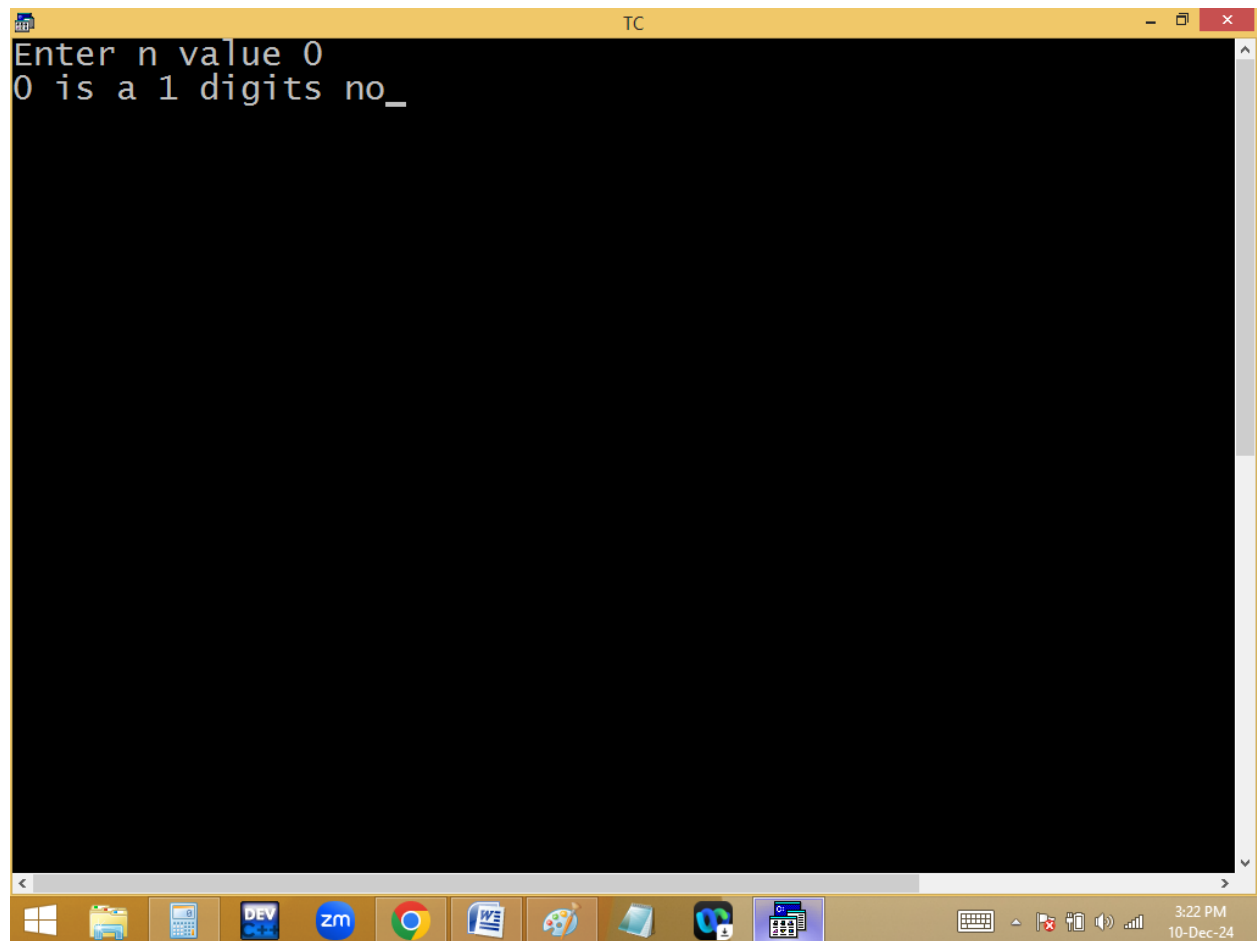
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Edit" menu, there is a submenu with the following options: Line 9, Col 12, Insert, Indent, Tab, Fill, Unindent, *, and E. The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int  c;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
c = printf("%ld",n);
if(n<0)c--;
printf(" is a %d digits no",c);
getch();
}
```

Below the code area, there is a status bar with the following text: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. At the bottom of the screen is a Windows taskbar with various icons, including the Start button, File Explorer, Calculator, DEV, ZM, Chrome, Word, Paint, and a folder icon. The system clock in the bottom right corner shows 3:22 PM on 10-Dec-24.







$$\frac{7}{143}$$

`c = printf("%d",n);`
`p("is a %d digits no",c);`

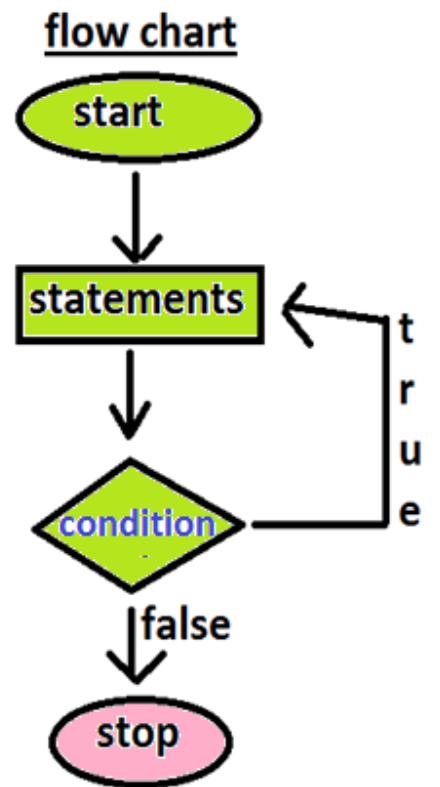
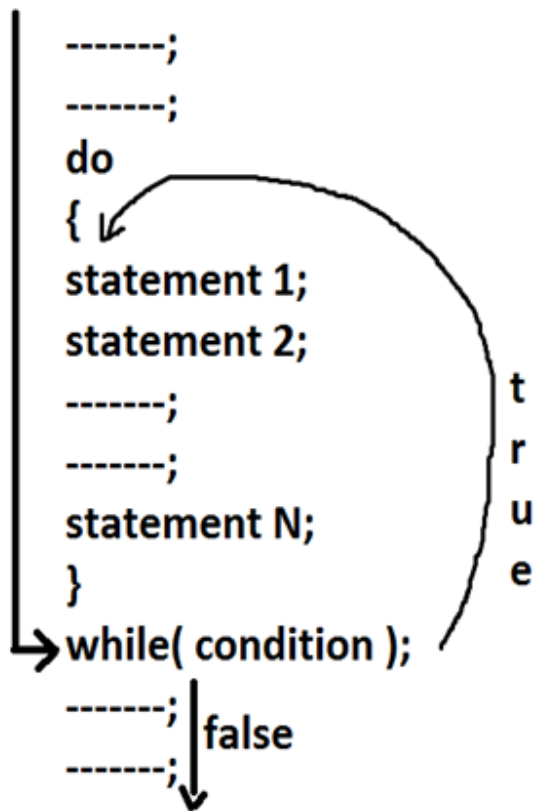
143
3
3
3

do..while:

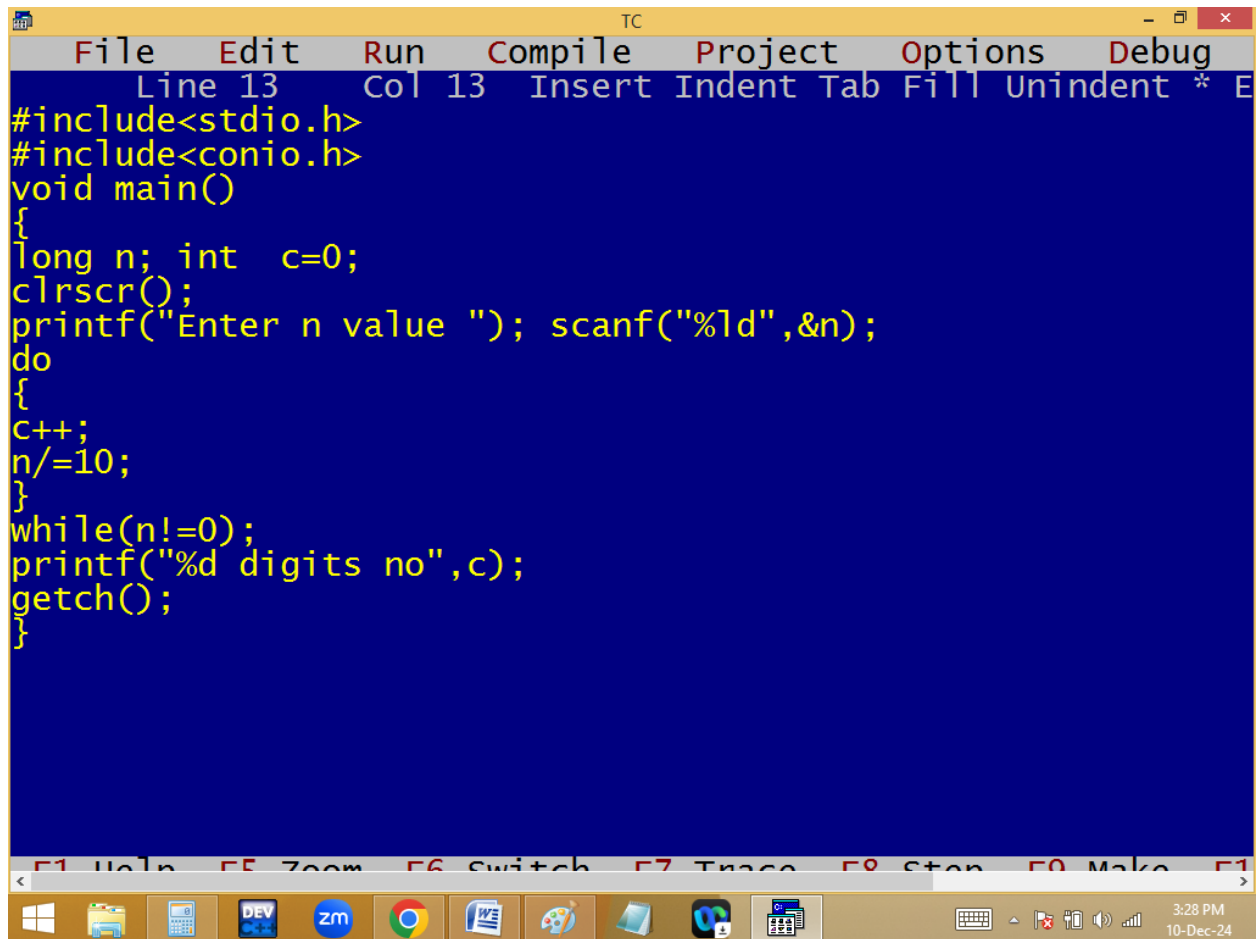
do .. while:

- It is an exit control loop. i.e. in a do while the condition is tested at last.
- Here do , while are the keywords.
- It is also used to repeat a program several times based on a condition.

- In a do while, do block statements are executed first and later while condition is tested. If the while condition is true then once again the do block statements are repeated. Like this the process is continued until the while condition becomes false.
- In do while, the while should be end with semicolon (;) .
- Regardless of while condition, the do statements are executed at least one time. Due to this sometimes we are getting unwanted results [garbage values].
- Use do while whenever it is compulsory because of in do while the program is controlled at the bottom / last.



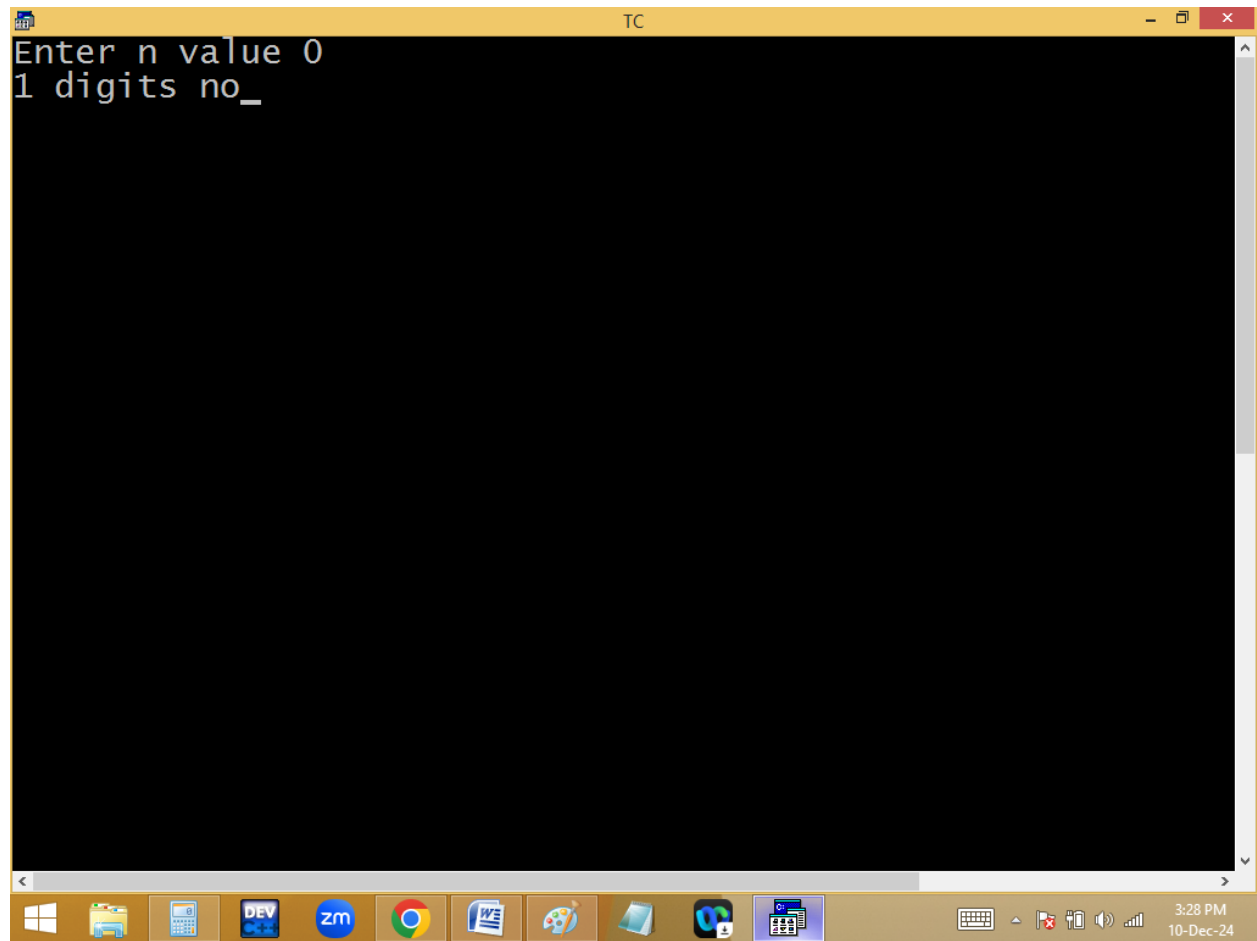
With loop:

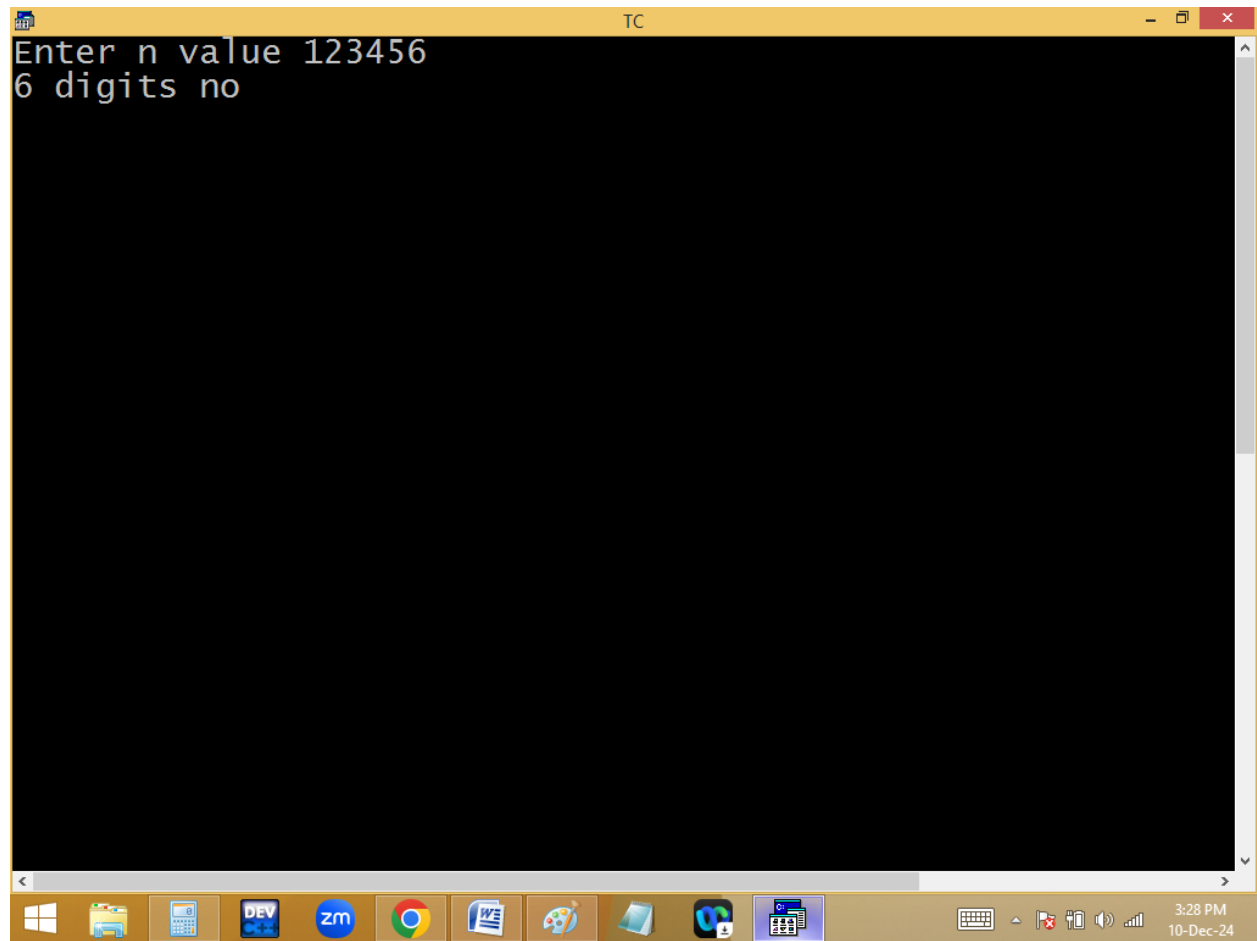


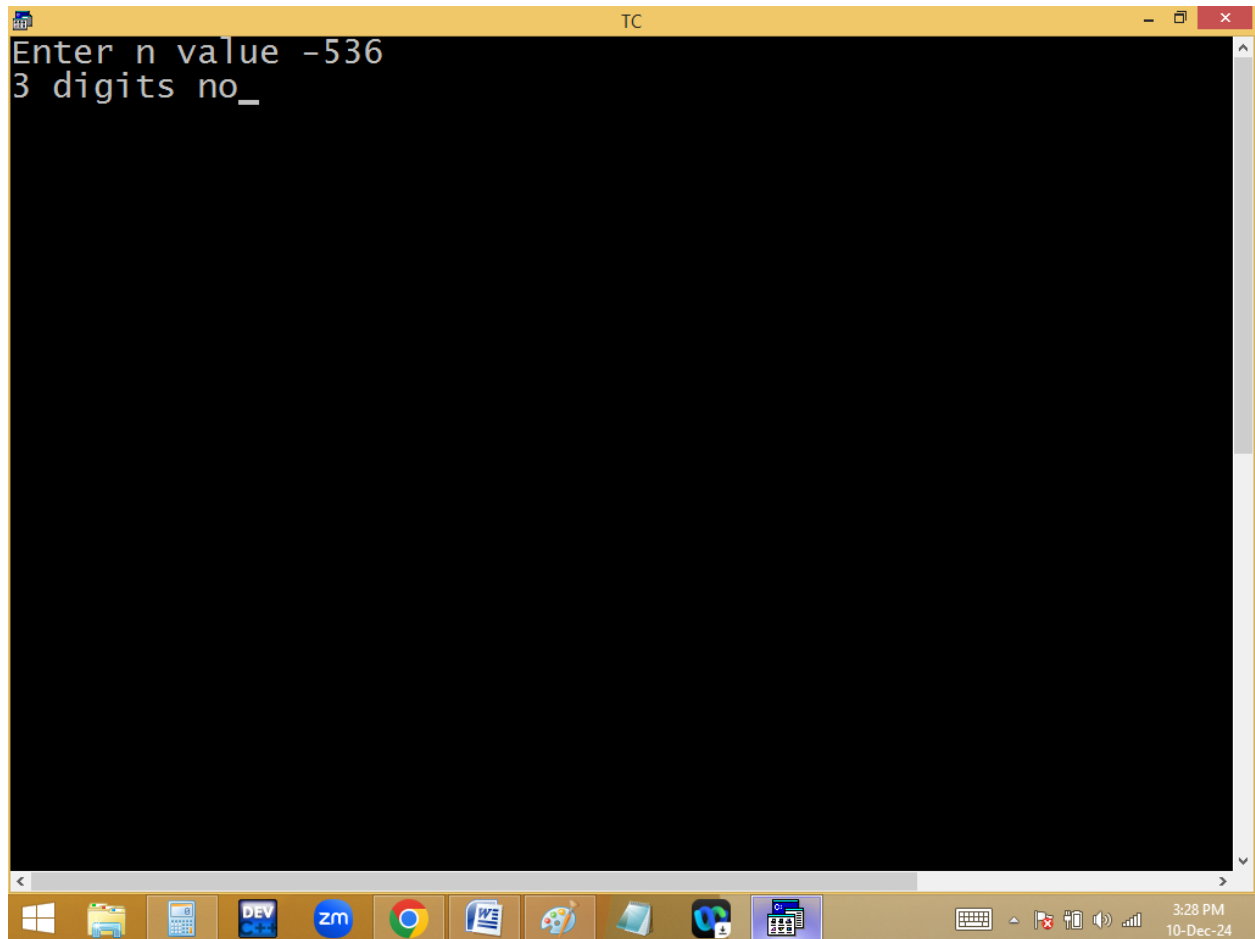
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 13 Col 13" and lists keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The code editor has a blue background and contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
long n; int c=0;
clrscr();
printf("Enter n value "); scanf("%ld",&n);
do
{
c++;
n/=10;
}
while(n!=0);
printf("%d digits no",c);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "3:28 PM" and "10-Dec-24".







The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text reads: "Enter n value -536" followed by "3 digits no_". The window has a yellow title bar with the text "TC" and standard Windows window controls. The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 3:28 PM on 10-Dec-24.

```
Enter n value -536
3 digits no_
```

Home work:

1. Finding first and last digits of given no.
Eg: 3047 → 7 last digit, 3 first digit.
2. Finding max, min digits of given no
Eg: 29317 → max=9, min=1
3. Finding the given digit position
Eg: 20589 → 0 position → 2nd